

B1GMB42 Software Inventory

Machine: Dell Precision Tower 5810

Hostname: Tower5810

OS: Ubuntu 26.04 LTS (resolute)

Companion doc: B1GMB42-slot-port-inventory.md (hardware)

Verified: 2026-07-03 — all listed packages present on this install

This document describes the **software stack** installed for ham radio, SDR, GNU Radio, HackRF/Mayhem, Python/Rust development, GPU testing, and Dell workstation tooling on the IndianaDell workspace.

Installation status (summary)

Layer	Status	Notes
Build / Python / docs	OK	cmake, python3-dev, pandoc, texlive-xetex
GNU Radio 3.10.12	OK	gr-osmosdr, satellites, dab, fosphor, ...
Ham radio desktop	OK	fldigi, WSJT-X, chirp, direwolf, xastir, ...
SDR hardware libs	OK	RTL-SDR, HackRF, Airspy, bladeRF, Lime, UHD, SoapySDR
HackRF / Mayhem v2.4.0	OK	firmware zips, SD payload, source repos, built tools
URH 2.10.0	OK	Python venv in hackrf/venv-urh/
Rust 1.96	OK	rustup stable in ~/.cargo/
GPU tools	OK	vulkan-tools, mesa-utils, clinfo
Flatpak	OK	Telegram 6.9.3
HackRF USB attached	—	not plugged in at last verify

Full apt lists: apt-hamradio-dev-manifest.txt (178 SDR/ham-related), apt-full-manifest.txt (generated by rebuild script).

Development toolchain

Component	Version / path	Purpose
Python	3.14	System + venvs
pip / venv	apt	Package isolation (URH)
numpy, scipy, matplotlib	apt	Science stack + GNU Radio
Rust	1.96.1	~/ .cargo/bin — source ~/ .cargo/env
cmake, gcc, clang	apt	Native + cross builds
ARM GCC	14.2 (apt)	PortaPack Mayhem firmware compile
pandoc + xelatex	apt	Manual PDF generation

Python bindings verified: gnuradio, SoapySDR, Hamlib (capital H).

GNU Radio & desktop SDR

Core: gnuradio, gnuradio-dev, gnuradio-doc — version **3.10.12.0**

Package	Role
gr-osmosdr	RTL-SDR, HackRF, generic OsmoSDR sources
gr-limesdr	LimeSDR blocks
gr-fosphor	GPU spectrum display
gr-air-modes	ADS-B / Mode S
gr-dab	DAB/DAB+ receiver
gr-satellites	Satellite telemetry decoders
gr-hpsdr	OpenHPSDR protocol
libgnuradio-soapy	SoapySDR integration

Application	Command	Use
GQRX	gqrx	General SDR receiver GUI
Quisk	quisk	SDR transceiver GUI
Flowgraphs	grcc file.grc	Compile GNU Radio Companion graphs
inspectrum	inspectrum file	Visualize captured IQ
URH	bin/urh	Reverse-engineer / replay protocols

SoapySDR modules: HackRF, RTL-SDR, Airspy, bladeRF, Lime, MiriSDR, HydraSDR, PlutoSDR, Red Pitaya, UHD, remote, audio.

Ham radio (desktop)

Application	Command	Role
fldigi	fldigi	Digital modes
WSJT-X	wsjtx	FT8, JT65, weak-signal
CHIRP	chirpw / chirpc	Radio programming
direwolf	direwolf	Sound-card TNC / APRS
gpredict	gpredict	Satellite pass prediction
grig	grig	Hamlib rig control GUI
xastir	xastir	APRS map client
Hamlib	python3-hamlib, libhamlib-dev	Rig control API

HackRF & PortaPack Mayhem

Recommended firmware: PortaPack Mayhem v2.4.0 — successor to Havoc; supports HackRF One, HackRF Pro, H4M.

Host tools (apt + built)

Tool	Source
hackrf_info, hackrf_transfer, hackrf_spiflash hackrf_sweep	apt + hackrf/build/ built from repos/hackrf
hacktv	apt — analog TV transmit
inspectrum	apt
dfu-util, openocd	apt — firmware programming
URH	hackrf/venv-urh/

Release assets (hackrf/releases/)

File	Size	Purpose
FIRMWARE_mayhem_v2.4.0.zip	2.4 MB	USB flash bundle
COPY_TO_SDCARD_hackrf_mayhem_v2.4.0	201 MB	PortaPack microSD content
no-world-map.zip		
OCI_hackrf_mayhem_v2.4.0.ppfw.tar	2.4 MB	Web flasher image

Extracted SD tree: `hackrf/sd-card/mayhem-v2.4.0/` (**276 MB**, 84 external apps in APPS/).

Mayhem firmware highlights (v2.4.0)

On-device: Morse RX/TX, RTTY, FPV detect, ADSB, ACARS, BLE, TPMS, KeeLoq, EPIRB, SAME, MDC-1200, P25, KISS TNC, Looking Glass, SubGHz, Flipper TX, waterfall designer, time sink.

SD apps include: `fpv_detect`, `kiss_tnc`, `keeloqtx`, `siggen`, `fmradio`, `sstvrx`, `wardrivemap`, `waterfall_designer`, and more.

Source repositories (`hackrf/repos/`)

`hackrf`, `mayhem-firmware` (+ submodules), `portapack-hackrf`, `urh`, `hacktv`

Themes (boot, login, desktop)

Module: Themes/ — each folder has a **README.md** explaining assets and how to modify them.

Stage	What you see	Apt packages	Customize
Boot	Dell BGRT center + spinner + Ubuntu watermark	<code>plymouth</code> , <code>plymouth-theme-spinner</code>	<code>boot/overlay/</code> then <code>sudo bin/themes-install-boot</code>
Login	GDM on GNOME Shell + Yaru-dark	<code>gdm3</code> , <code>gnome-shell</code> , <code>yaru-theme-gnome-shell</code>	<code>bin/apply-dark-mode</code>
Desktop	Yaru-dark GTK + icons + shell	<code>yaru-theme-gtk</code> , <code>yaru-theme-icon</code> , ...	<code>bin/apply-dark-mode</code>

Extracted boot logos: Themes/boot/extracted/bgrrt-firmware-oem.png (Dell), ubuntu-watermark-dark.png (bottom).

```
bin/themes-extract           # refresh mirrors (~193 MB) + extract logos
sudo bin/themes-install-boot  # custom Plymouth (reboot to preview)
sudo bin/themes-restore-boot  # factory BGRT theme
bin/apply-dark-mode          # dark GTK/shell/GDM
```

Start at Themes/README.md, then boot/, login/, or desktop/ READMEs.

GPU & machine tooling

Tool	Launcher	Role
GPU stress	bin/gpu-stress	3× FirePro Vulkan test (vkcube)
IO benchmark	bin/iotest	Disk survey + sequential RW
Dell inventory	bin/dellmerge	System report for *.report files
AMDGPU config	bin/apply-amdgpu	Install etc/ multi-GPU configs (sudo)
AMD ROCm install	bin/amd-install	Optional — amd-radeon/ scripts
GPU verify	vkcube, clinfo, glxinfo	apt packages

GPUs: 2× FirePro W5000 + 1× W5100 — Vulkan/OpenCL OK; ROCm ML not supported on these cards.

IndianaDell workspace

```
IndianaDell/
├── B1GMB42-slot-port-inventory.md    # hardware manual
├── B1GMB42-software-inventory.md    # this file
├── apt-hamradio-dev-manifest.txt    # SDR/ham apt subset
├── apt-full-manifest.txt            # full dpkg list (after rebuild)
├── bin/                             # launchers (see table below)
├── scripts/
│   ├── dell/, gpu/, storage/, gnome/, rebuild/
│   └── Themes/                     # boot/login/desktop -
└── README in each folder
    ├── boot/    # Plymouth, BGRT Dell logo, ubuntu watermark
    ├── login/   # GDM greeter mirrors
    └── desktop/ # Yaru GTK/icon/shell mirrors
├── hackrf/      # Mayhem, URH, repos, builds
├── amd-radeon/  # optional ROCm installer
├── etc/         # multi-GPU Wayland/X11 config
├── docs/features-available.md    # quick reference
└── FactoryDocs/                 # Dell driver packages
```

bin/ launchers

Command	Runs
bin/rebuild-machine	Full software stack restore
bin/dellmerge	Dell workstation inventory
bin/gpu-stress	GPU stress test
bin/iotest	Storage benchmark
bin/apply-amdgpu	Apply etc/ GPU configs
bin/amd-install	ROCm driver install
bin/hackrf-env	Source HackRF PATH (source, not exec)
bin/urh	Universal Radio Hacker
bin/hackrf-flash-mayhem	Extract Mayhem flash bundle
bin/hackrf-prepare-sdcard	Extract SD card payload
bin/hackrf-build-mayhem	Compile Mayhem from source
bin/hackrf-setup-udev	HackRF USB permissions
bin/themes-extract	Snapshot theme apt packages + boot logos
bin/themes-install-boot	Custom Plymouth splash (sudo)
bin/themes-restore-boot	Stock boot splash (sudo)
bin/apply-dark-mode	GNOME dark (session + GDM)
bin/apply-max-performance	No suspend / dim / night-light

Flatpak

App	ID	Version
Telegram	org.telegram.desktop	6.9.3

Rebuild this machine from scratch

After a fresh Ubuntu 26.04 install (with IndianaDell copied or cloned to ~/Documents/IndianaDell):

```
cd ~/Documents/IndianaDell
chmod +x bin/* scripts/rebuild/*.sh
bin/rebuild-machine           # ~15-30 min (downloads Mayhem, builds HackRF, etc)
bin/rebuild-machine --verify-only # check only, no installs
```

What the rebuild script does:

1. apt install — all core, dev, GNU Radio, ham, SDR, HackRF packages
2. Flatpak + Telegram (skip: SKIP_TELEGRAM=1)

3. rustup stable (if missing)
4. Clone HackRF/Mayhem/URH repos; init Mayhem submodules
5. Build HackRF host tools (skip: SKIP_HACKRF_BUILD=1)
6. Download Mayhem v2.4.0 + extract SD card tree
7. Create URH venv
8. Install HackRF udev rules
9. Regenerate apt-full-manifest.txt and apt-hamradio-dev-manifest.txt
10. Verify all components

What it does not do: disk partitioning, ZFS, BIOS, Windows, AMD etc/ deploy, ROCm install, or HackRF hardware flash. Run those separately:

```
sudo bin/apply-amdgpu && sudo reboot      # GPU session config
bin/amd-install                          # optional ROCm
bin/hackrf-flash-mayhem                  # when HackRF is plugged in
```

Quick verification

```
cd ~/Documents/IndianaDell
bin/rebuild-machine --verify-only
source bin/hackrf-env
hackrf_info                               # needs hardware
gnuradio-config-info --version
. ~/.cargo/env && rustc --version
bin/urh --version
```

Not installed / known gaps

Item	Notes
SDRangel, SigDigger	Not on Flathub; use gqrx + URH + inspectrum
Rust SDR crate workspace	Toolchain only — add crates per project
Pre-crash apt list	Lost in TPM/ZFS crash; use current manifests
HackRF hardware test	No device attached at last verify
ROCM / HIP ML	W5000/W5100 not in AMD ROCm matrix

Refresh commands

```
cd ~/Documents/IndianaDell
bin/rebuild-machine --verify-only
pandoc B1GMB42-software-inventory.md -o B1GMB42-software-inventory.pdf \
  --pdf-engine=xelatex -V mainfont="Noto Sans" -V monofont="DejaVu Sans Mono"
```

IndianaDell software inventory. Last updated: 2026-07-03.